

IVIVC App User Guide

Contents

1. What the app does	2
2. Input datasets	2
3. Analysis approaches and models	2
4. Preprocessing options	3
Normalization	3
Scaling	3
Interpolation	3
5. Relative model evidence	3
6. Cross-validation options	4
7. Grid-search initialization	5
8. Performance metrics	6
9. Uncertainty and error bars	7
Approach 1: Direct Correlation	7
Approach 2: Independent Model Fitting	8
Approach 3: Direct Mapping	8
10. Outputs	9
11. Command-line interface	9
12. Regenerating this guide	9

Appendix: How to Interpret Model Performance **9**

1. Confirm that models are comparable before interpreting metrics	10
2. Metric ranges, direction, and interpretation	10
3. Interpreting goodness of fit	11
4. Interpreting prediction error	11
5. Interpreting relative model evidence	11
6. Interpreting cross-validation performance	12
7. Inspecting residuals and diagnostic plots	12
8. Interpreting parameter uncertainty	13
9. Considering model complexity	13
10. Scientific plausibility and intended use	13
11. Suggested model-selection workflow	13
12. Example interpretation statements	14
13. Limitations	14
References and suggested reading	14

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable polymers by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. IVIVCs may help evaluate manufacturing or material changes, support predictions of in vivo performance when modifying an existing device design, and reduce reliance on animal studies during preclinical development. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data, and may optionally supply an in vitro prediction dataset. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries that can be reviewed and incorporated into external documentation.

1. What the app does

- Apply selected normalization, scaling, and interpolation options to prepare fitting and prediction data.
- Fit user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculate selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generate fitting, comparison, and prediction plots with uncertainty displays where supported.
- Create report-ready HTML and editable Word summaries for user-selected final models or methods.

2. Input datasets

Input	Description
Fitting dataset	Required four-column dataset used to establish the in vitro/in vivo relationship. Columns are interpreted as in vitro time, in vitro value, in vivo time, and in vivo value.
Prediction dataset	Optional two-column in vitro dataset used after fitting to generate in vivo predictions. Columns are interpreted as in vitro time and in vitro value.

For Excel workbooks, select the sheet containing the relevant data. For CSV files, the first columns are read directly in the order described above.

3. Analysis approaches and models

Approach	Description	Available models
Approach 1: Direct Correlation	Approach 1 fits a direct relationship between matched in vitro and in vivo time points after preprocessing. The fitted model is then used as a time-scaling function that maps an in vitro time to a predicted in vivo time.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power
Approach 2: Independent Model Fitting	Approach 2 fits the same model family independently to the in vitro and in vivo datasets. With fitting data alone, the report summarizes the two fitted models and their fitted-curve uncertainty. When prediction data are supplied, predictions can be generated by comparing the fitted in vivo and in vitro responses or by comparing fitted characteristic time constants when those values are available.	Double Exponential, Exponential, Mass Loss (Gompertz), Mass Loss (One-Phase), Mass Loss (Surface Erosion), Mass Loss (Two-Phase), Power, Stretched Exponential
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. It can describe value-ratio behavior over time and time-ratio behavior over value ranges. When prediction data are supplied, those mappings can be used to scale prediction values or prediction times.	No parametric model selection

4. Preprocessing options

Normalization

Option	Internal key	Description
Min-max normalization	<code>min_max</code>	Scales response values to the 0-1 range using the observed minimum and maximum within each dataset.
Initial-value normalization	<code>initial_value</code>	Divides response values by the first response value in the corresponding dataset.

Scaling

Option	Internal key	Description
Log10 time scaling	<code>log_x</code>	Applies a base-10 logarithm to time values before fitting or interpolation.
Log10 response scaling	<code>log_y</code>	Applies a base-10 logarithm to response values before fitting or interpolation.

Interpolation

Option	Internal key	Description
No interpolation	<code>none</code>	Uses only exact raw alignments. No values are estimated between observations; unavailable aligned positions are marked as missing.
Shared-time interpolation	<code>default</code>	Interpolates both datasets onto the common set of experimental time points, using linear interpolation and marking out-of-range values as unavailable.
Densified interpolation	<code>alternative</code>	Creates additional linearly spaced points between adjacent observations before interpolation. This can smooth displays and make fitting more stable, but adds more interpolation-derived points that skew statistical comparisons.

5. Relative model evidence

Relative model evidence compares candidate parametric models fitted to the same dataset under the same preprocessing settings and IVIVC approach. The evidence table is based only on the information-criterion metrics selected by the user: AIC, AICc, and/or BIC. These criteria are relative; their absolute values are not interpreted directly, and negative values are allowed. Table definitions: *n* is the number of fitted observations; *k* is the number of fitted model parameters; *Delta* is the difference between a model's criterion value and the lowest value in the displayed candidate set; *weight* is the normalized relative support for that model under the criterion; and *evidence ratio* compares the highest weight in the set with the model's weight. *Delta* values near 0 indicate similar relative support; larger *delta* values indicate lower support relative to the leading candidate under that criterion. Similar relative evidence does not mean the models are scientifically equivalent. Final model selection should also consider prediction error,

cross-validation behavior, residual patterns, parameter uncertainty, model simplicity, and biological or degradation-mechanism plausibility. The app reports these summaries to support user review. It does not select, recommend, or validate a model.

Evidence summary	Interpretation
Delta AICc / Delta AIC / Delta BIC	Difference from the lowest value in the displayed candidate set. Values near 0 indicate similar relative support; larger values indicate lower support relative to the leading candidate under that criterion.
AICc or BIC weight	Normalized relative-support value across the candidate models in the displayed evidence table. Weights are relative, not absolute probabilities that a model is true.
Evidence ratio	Ratio of the highest evidence weight to the model's evidence weight. Larger ratios indicate less relative support compared with the highest-weighted model in the set.
Interpretation badge	Neutral rule-of-thumb label based on the displayed delta values. The label supports user review and is not an app recommendation.

6. Cross-validation options

Cross-validation provides an approximate check of how a fitted model performs on held-out data. It is most useful when there are enough observations to fit the model repeatedly while leaving some observations out for evaluation.

The available schemes are Shuffle split, K-fold, Leave-one-out, and None. Shuffle split uses repeated random holdout sets, K-fold uses systematic fold-based holdouts, Leave-one-out holds out one observation at a time, and None disables cross-validation. For small IVIVC datasets, cross-validation summaries should be interpreted as exploratory diagnostics rather than definitive validation results.

Scheme	Internal key	Description
Shuffle split	<code>shuffle_split</code>	Repeatedly creates random training and holdout subsets. This can be useful when the dataset has enough observations to support repeated resampling. The test-size parameter controls the fraction held out in each split, and the random seed makes the random splits reproducible.
K-fold	<code>kfold</code>	Divides the dataset into a specified number of folds, fits the model on all but one fold, and evaluates on the held-out fold. Each fold is used once as the holdout set. The splits/folds parameter controls the number of folds. The data are randomly shuffled before splitting into folds (e.g., to avoid folds with only short-time data being used to predict long-time data), and the random seed makes the shuffling reproducible.

Scheme	Internal key	Description
Leave-one-out	<code>leave_one_out</code>	Uses one observation as the holdout set and fits the model on all remaining observations, repeating until each observation has been held out once. This can be useful for very small datasets, but the results can be variable and computationally slower.
None	<code>none</code>	Disables cross-validation. This may be appropriate when the dataset is too small for meaningful holdout validation or when only fitted-model summaries are needed.

Parameter	Internal key	Description
Cross-validation scheme	<code>cv_scheme</code>	Selects the splitting method used to estimate held-out performance for fitted parametric models.
CV splits/folds	<code>cv_n_splits</code>	For Shuffle split, the number of repeated random train/test splits. For K-fold, the number of folds. This setting is not used for Leave-one-out or None.
ShuffleSplit test size	<code>cv_test_size</code>	The fraction of observations held out for testing in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the holdout test more demanding but leave fewer observations for training.
CV random seed	<code>cv_random_state</code>	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split and shuffled K-fold.

7. Grid-search initialization

The app conducts an automated grid search to identify suitable starting parameter values for each model before nonlinear least-squares fitting, reducing the need for manual parameter tuning. Candidate values are sampled within the configured parameter bounds and are used only to initialize the final curve fit.

This initialization step improves convergence and reduces sensitivity to poor starting values. It is not itself the final fitting algorithm and does not guarantee that the final fitted parameters are the global optimum.

Parameter	Internal key	Description
Grid-search random starts	<code>grid_search_num_points</code>	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.
Grid-search CPU cores	<code>grid_search_num_cores</code>	Number of CPU cores used for the starting-point search. Blank or Auto uses the app default.

Parameter	Internal key	Description
Grid-search parameter minimum	<code>grid_search_param_min</code>	Lower bound for randomly sampled candidate starting parameter values.
Grid-search parameter maximum	<code>grid_search_param_max</code>	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	<code>grid_search_random_state</code>	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

8. Performance metrics

Metric	Internal key	Preferred direction	Description
R ²	<code>r_squared</code>	Higher is generally better	Coefficient of determination. Higher values indicate that the model explains more variance in the observed response.
Adjusted R ²	<code>adjusted_r_squared</code>	Higher is generally better	R ² adjusted for the number of fitted parameters. Higher values are better and can penalize unnecessary model complexity.
MSE	<code>mse</code>	Lower is generally better	Mean squared error between observed and fitted values. Lower values indicate smaller average squared residuals.
RMSE	<code>rmse</code>	Lower is generally better	Root mean squared error, in the same response units as the fitted data. Lower values indicate smaller typical residuals.
NRMSE (mean)	<code>mnmse</code>	Lower is generally better	RMSE normalized by the response mean. Lower values indicate smaller error relative to the response magnitude.
NRMSE (std. dev.)	<code>nrmse</code>	Lower is generally better	RMSE normalized by the response standard deviation. Lower values indicate smaller error relative to response variability.

Metric	Internal key	Preferred direction	Description
AIC	aic	Lower is generally better	Akaike Information Criterion. Lower values indicate stronger relative fit-complexity support among models fitted to the same dataset under comparable assumptions. Absolute AIC values, including negative values, should not be interpreted as standalone quality grades.
AICc	aicc	Lower is generally better	Corrected Akaike Information Criterion. AICc applies a small-sample correction to AIC and is often more appropriate when the number of fitted observations is small relative to the number of fitted parameters. Lower values indicate stronger relative fit-complexity support among comparable fitted models; AICc is undefined when there are too few observations relative to the number of parameters.
BIC	bic	Lower is generally better	Bayesian Information Criterion. Lower values indicate stronger relative fit-complexity support among models fitted to the same dataset under comparable assumptions, with a stronger penalty for additional parameters than AIC. Absolute BIC values, including negative values, should not be interpreted as standalone quality grades.

9. Uncertainty and error bars

The following descriptions summarize how fitted-parameter uncertainty, fitting-plot bands, and prediction error bars are calculated or interpreted for each approach.

Approach 1: Direct Correlation

Fitting uncertainty

Parameter estimates: fitted parameters are reported as estimate $\pm$ standard error. Standard errors are derived from the nonlinear least-squares parameter covariance matrix. Correlations among fitted parameters are retained

when uncertainty-aware parameter values are created.

Fitting plot bands: shaded 95% prediction bands are generated by Monte Carlo sampling of the fitted parameter covariance matrix. The sampled model variance is combined with the residual variance from the fitted data, then converted to a two-sided interval using a t multiplier when residual degrees of freedom are available.

These bands describe approximate prediction uncertainty around the fitted relationship, not regulatory acceptance limits. They depend on the fitted model form, local covariance approximation, residual behavior, and preprocessing choices.

Prediction uncertainty

Prediction plot error bars: when a prediction dataset is supplied, predicted in vivo times are plotted with horizontal error bars. The bars are calculated from the same 95% prediction-band calculation used for the fitted correlation model; the upper and lower bar lengths are the distances from the band mean to the corresponding upper and lower interval bounds at each input prediction time.

The error bars therefore include propagated fitted-parameter uncertainty and residual/model error from the Approach 1 correlation fit.

Approach 2: Independent Model Fitting

Fitting uncertainty

Parameter estimates: each in vitro and in vivo model parameter is reported as estimate $\pm$ standard error. Standard errors are derived from the corresponding nonlinear least-squares covariance matrix, with fitted-parameter correlations retained in uncertainty-aware values.

Time constants: tau values, when available, are calculated from the fitted model and parameter covariance. The reported tau uncertainty is the propagated standard uncertainty from that covariance-based calculation.

Fitting plot bands: the in vitro and in vivo shaded 95% prediction bands are generated separately. For each fitted model, the app Monte Carlo samples the fitted parameter covariance matrix, combines sampled model variance with residual variance, and converts the resulting standard uncertainty to a two-sided interval using a t multiplier when residual degrees of freedom are available.

Prediction uncertainty

Value-ratio prediction error bars: uncertainty is propagated from both fitted models. The app samples each model's parameter covariance matrix, calculates the in vivo/in vitro response ratio for each sample, combines ratio variance with residual-based ratio uncertainty, and converts that uncertainty to a two-sided 95% interval. The plotted vertical error bars are scaled from the resulting ratio interval at each prediction time.

Time-constant-ratio prediction error bars: when both tau values are valid, uncertainty in each tau is propagated through the tau ratio using uncertainty-aware arithmetic. The standard uncertainty of the scaled prediction time is multiplied by a t value, where available, and plotted as a symmetric horizontal error bar.

Ratio-based uncertainty can become unstable when the denominator model prediction or denominator time constant is near zero. Predictions in those regions should be interpreted cautiously.

Approach 3: Direct Mapping

Fitting uncertainty

Parameter uncertainty: Approach 3 does not fit a parametric model, so there is no fitted-parameter covariance matrix, no parameter standard errors, and no model-based 95% prediction band.

Fitting plots: the plots show interpolation-based ratios/mappings calculated from the preprocessed datasets. They should be interpreted as deterministic mappings from the supplied data and preprocessing choices, not as statistical uncertainty bands.

Prediction uncertainty

Prediction plot error bars: Approach 3 currently does not generate formal propagated error bars for prediction outputs because no parametric covariance model is fitted.

Uncertainty is therefore limited to the quality of the input data, preprocessing, interpolation, and the assumption that the observed mapping remains valid for the prediction dataset. Extrapolated points outside the interpolation range are excluded or returned as unavailable.

10. Outputs

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction.
Model fitting outputs	Parameter estimates, parameter standard errors where available, fitted plots, and selected performance metrics.
Model comparison outputs	Tables comparing selected approaches, models, metrics, and cross-validation summaries.
Prediction outputs	Prediction plots and downloadable tables when a prediction dataset is supplied and the selected method can be applied.
Selected model & summary of outputs	User-selected report summaries and optional Word report export for incorporation into external documentation.

Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

11. Command-line interface

The repository includes `cli.py` for batch-oriented fitting from the command line. Use `python cli.py --help` to see available arguments for input files, approaches, models, metrics, preprocessing options, cross-validation settings, and grid-search settings. The internal keys specified in this user guide are used as settings with `cli.py`.

12. Regenerating this guide

Most descriptive text in this guide is generated from `utilities/descriptions.py`, with routine wording overrides in `utilities/descriptions.yaml`. Long-form appendices are sourced from Markdown files in `docs/appendices/`. After editing those files, regenerate the Markdown files with:

```
python utilities/generate_user_guide.py
```

To generate a PDF, use a Markdown-to-PDF tool such as Pandoc:

```
pandoc docs/user_guide.md -o docs/user_guide.pdf
```

Appendix: How to Interpret Model Performance

No single performance metric determines whether an IVIVC model is meaningful. Model selection should consider goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the polymer degradation mechanism and the intended context of use.

The app reports metrics and comparison tables to support user review. It does not select, recommend, validate, or establish regulatory acceptability for any model. Users should document the rationale for the final model or method selected for a specific analysis.

1. Confirm that models are comparable before interpreting metrics

Model-performance metrics are most useful when the compared models were evaluated under the same analysis conditions. In general, compare models only within the same uploaded dataset, preprocessing configuration, response scale, selected IVIVC approach, and candidate-model set.

Do not directly compare AIC, AICc, or BIC values across different datasets, transformed responses, preprocessing settings, or substantially different fitting contexts. These information criteria are relative comparison tools, not absolute quality scores. Approach 3 should also be interpreted differently from Approaches 1 and 2 because it uses direct interpolation-based mappings rather than fitting a parametric model.

2. Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. R^2 can be negative when the model performs worse than the mean-response baseline. Higher is generally better.	R^2 summarizes the fraction of response variance explained by the fitted model. High R^2 does not prove that the model is mechanistic, causal, predictive outside the fitted data range, or appropriate for the intended use.
Adjusted R^2	Higher is generally better. It can be lower than R^2 and may be negative.	Adjusted R^2 penalizes additional fitted parameters and may be more informative than R^2 when comparing models with different complexity.
MSE	Lower is generally better. Units are the squared response units.	MSE summarizes average squared error. It is useful computationally but can be less intuitive than RMSE because the units are squared.
RMSE	Lower is generally better. Units match the response units.	RMSE summarizes typical prediction error on the response scale. Interpret RMSE relative to assay variability, measurement uncertainty, clinically or biologically meaningful differences, and the intended use.
NRMSE	Lower is generally better. The value is normalized and is intended to be more comparable across response scales.	NRMSE expresses prediction error relative to a normalization quantity, such as the response mean or response standard deviation. Interpretation depends on the normalization method.
AIC	Lower is generally better, but only relative to other comparable models. AIC has no fixed lower or upper bound, and negative values are allowed.	AIC balances fitted error and model complexity. Absolute AIC values should not be interpreted as standalone quality grades.
AICc	Lower is generally better, but only relative to other comparable models. AICc is undefined when there are too few observations relative to the number of fitted parameters.	AICc is AIC with a small-sample correction. It is often more appropriate than AIC when the number of observations is limited relative to the number of parameters.

Metric	Range / direction	Interpretation guidance
BIC	Lower is generally better, but only relative to other comparable models. BIC has no fixed lower or upper bound, and negative values are allowed.	BIC also balances fitted error and model complexity, but it usually penalizes additional parameters more strongly than AIC.
Cross-validation metrics	Direction follows the underlying metric: higher is generally better for R^2 -type metrics, and lower is generally better for error metrics.	Cross-validation metrics estimate held-out performance under the selected validation scheme. They can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted-data and cross-validation performance.	For models with consistent fitted and cross-validation performance, the fit/CV ratio is expected to be near 1. Substantial deviations from 1 may indicate reduced generalizability, overfitting, model instability, or sensitivity to the validation procedure, although no universal acceptance threshold is established.

3. Interpreting goodness of fit

Goodness-of-fit metrics describe how well the model fits the data used to estimate the model parameters. These metrics include R^2 , adjusted R^2 , MSE, RMSE, and NRMSE.

High fitted-data performance can be encouraging, but it can also be optimistic, especially for small datasets, flexible models, or models with many parameters. A model with strong fitted-data performance may still be unsuitable if it has unstable parameters, poor cross-validation performance, systematic residual patterns, or a fitted shape that is not scientifically plausible for the degradation process.

When interpreting fitted-data metrics, consider whether the residual errors are small enough to be meaningful for the specific degradation endpoint and context of use. There is no universal RMSE or NRMSE cutoff for IVIVC adequacy.

4. Interpreting prediction error

Prediction-error metrics should be interpreted in practical context. RMSE is often easier to interpret than MSE because it is reported in the same units as the response. NRMSE can be useful for comparing error across response scales, but the denominator matters. For example, NRMSE normalized by the response mean and NRMSE normalized by the response standard deviation answer different questions.

A prediction error may be numerically small but practically important if the relevant assay, device attribute, or degradation endpoint has narrow acceptable variation. Conversely, a larger numerical error may be acceptable for exploratory model development if the intended use does not require high predictive precision.

5. Interpreting relative model evidence

Relative Model Evidence tables summarize information-criterion comparisons among candidate parametric models fitted under comparable conditions. These tables may include AIC, AICc, BIC, delta values, evidence weights, and evidence ratios.

Evidence quantity	Interpretation
AIC, AICc, or BIC	Information criterion value. Lower values indicate stronger relative support within the displayed candidate set. The absolute value is not interpreted directly.
Delta AIC, Delta AICc, or Delta BIC	Difference between a model's criterion value and the lowest value in the displayed candidate set. Values near 0 indicate similar relative support. Larger values indicate lower relative support under that criterion.
Evidence weight	Normalized relative-support value across the displayed candidate models. Weights are comparative summaries, not absolute probabilities that a model is true.
Evidence ratio	Ratio of the highest evidence weight to the model's evidence weight. Larger values indicate greater separation from the highest-weighted model under that criterion.

Common rules of thumb for AIC-family comparisons are: delta values from 0 to 2 suggest similar or substantial empirical support; values from 4 to 7 suggest considerably less support; and values greater than 10 suggest essentially no support relative to the leading candidate in that model set. Common rules of thumb for BIC-family comparisons are: delta values from 0 to 2 suggest weak differences; 2 to 6 suggest positive evidence against the higher-BIC model; 6 to 10 suggest strong evidence; and values greater than 10 suggest very strong evidence. These are interpretive conventions, not regulatory acceptance criteria.

Relative evidence does not prove that a model is correct, mechanistic, predictive outside the observed range, or suitable for a regulatory submission. If multiple models have similar relative support, users should also consider simplicity, parameter plausibility, residual behavior, cross-validation behavior, and scientific interpretability.

6. Interpreting cross-validation performance

Cross-validation estimates how a fitted model performs on held-out data. It is useful for assessing whether fitted-data performance appears to generalize beyond the observations used for parameter estimation.

Cross-validation results depend on the selected validation method, number of folds or splits, random seed, test-set size, sample size, and data distribution. For small IVIVC datasets, cross-validation should usually be treated as an exploratory diagnostic rather than a definitive validation result. Leave-one-out cross-validation can be useful when datasets are small, but it may be sensitive to individual observations. K-fold and shuffle-split methods can provide repeated or fold-based estimates, but they require enough observations to create meaningful training and holdout sets.

Fit/CV ratios summarize the relationship between fitted-data and held-out performance for the same metric. Ratios near 1 indicate that fitted and cross-validation scores are similar. Ratios far from 1 may suggest reduced generalizability, overfitting, model instability, or sensitivity to the validation scheme. The direction of concern depends on the metric and context, so the ratio should be interpreted together with the raw fit and CV values.

AIC, AICc, and BIC should not be interpreted as ordinary cross-validation scores because the final model and the cross-validation models are fitted to different subsets of the data and may involve different effective sample sizes. Use information criteria for relative evidence among comparable fitted models, and use cross-validation metrics to evaluate held-out predictive behavior.

7. Inspecting residuals and diagnostic plots

Summary metrics can hide important model behavior. Residual plots should be reviewed for systematic patterns, bias, curvature, clustering, changing variance, or influential observations.

Residual pattern	Possible implication
Random scatter around zero	More consistent with an adequate model form, although not proof of validity.

Residual pattern	Possible implication
Curvature or time-dependent trend	The model form may not capture the observed relationship.
Increasing or decreasing spread	Error variability may change across time, response, or degradation range.
Mostly positive or mostly negative residuals	Potential systematic bias.
One point dominates the apparent fit	The model may be sensitive to individual observations or limited data density.

Outliers should be investigated using scientific and data-quality rationale. Removing observations solely to improve metrics can make the model-selection rationale difficult to justify.

8. Interpreting parameter uncertainty

Parameter estimates should be reviewed for scientific plausibility and numerical stability. Large standard errors, wide confidence intervals, failed covariance estimation, or highly correlated parameters may indicate that the data do not strongly identify the model parameters.

Parameter uncertainty is especially important when a fitted model will be interpreted mechanistically, used to support extrapolation, or used to compare degradation behavior across materials, processes, or test conditions. A model with strong fitted metrics but poorly estimated or implausible parameters may be difficult to justify.

9. Considering model complexity

More complex models can often fit observed data more closely, but they may overfit small datasets or produce unstable parameter estimates. Adjusted R^2 , AIC, AICc, BIC, cross-validation metrics, and residual diagnostics can help evaluate whether additional complexity is justified.

When two models have similar fitted performance and similar relative evidence, the simpler and more interpretable model may be easier to justify, provided it is scientifically plausible and performs adequately for the intended use. Model complexity should be supported by data density, degradation-mechanism expectations, and the intended context of use.

10. Scientific plausibility and intended use

Statistical performance should be interpreted together with knowledge of the polymer, degradation mechanism, test conditions, and proposed use of the IVIVC. A model may be statistically competitive but scientifically difficult to justify if its fitted shape, parameter values, or extrapolated behavior conflict with known degradation behavior.

The intended use should determine the amount and type of supporting evidence needed. Exploratory model development may tolerate more uncertainty than a model intended to support a specific regulatory argument, design change, or prediction of in vivo performance.

11. Suggested model-selection workflow

1. Confirm that the dataset, preprocessing choices, and selected response variables are appropriate.
2. Review fitted curves and residual plots for each candidate model.
3. Compare fitted-data metrics, including R^2 , adjusted R^2 , RMSE, and NRMSE.
4. Review prediction-error metrics in the context of assay variability, endpoint importance, and intended use.
5. Review cross-validation performance and fit/CV ratios, where cross-validation is applicable.
6. Review Relative Model Evidence using AIC, AICc, and/or BIC, where parametric model comparisons are applicable.
7. Review fitted-parameter estimates, standard errors, covariance behavior, and scientific plausibility.
8. Consider model complexity and whether additional parameters are justified by the data and degradation mechanism.
9. Select the model or method based on a documented balance of statistical performance, diagnostic behavior, scientific plausibility, and intended use.
10. Avoid presenting the selected model as automatically chosen, validated, or accepted by the app.

12. Example interpretation statements

The following examples illustrate neutral wording that may be adapted for reports. Users should tailor the language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating strong fitted-data performance under the selected preprocessing conditions.
- Model B had similar AICc support to Model A, with Delta AICc less than 2, suggesting that the two models were not clearly distinguished by AICc within this candidate set.
- Although Model C had strong fitted-data performance, its cross-validation error was substantially higher than its fitted error, suggesting possible sensitivity to the validation procedure or reduced generalizability.
- The selected model was chosen based on a balance of fitted performance, cross-validation behavior, relative evidence, parameter plausibility, residual diagnostics, and scientific interpretability.

13. Limitations

Model-performance metrics are descriptive and comparative tools, not automatic acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. Users should document preprocessing assumptions, fitting settings, model-selection rationale, limitations, and any planned use of the model outside the fitted data range.

References and suggested reading

1. U.S. Food and Drug Administration. *Assessing the Credibility of Computational Modeling and Simulation in Medical Device Submissions: Guidance for Industry and Food and Drug Administration Staff*. November 2023. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions>
2. U.S. Food and Drug Administration. *Reporting of Computational Modeling Studies in Medical Device Submissions: Guidance for Industry and Food and Drug Administration Staff*. September 2016. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/reporting-computational-modeling-studies-medical-device-submissions>
3. Burnham, K. P., and Anderson, D. R. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. Springer, 2002.
4. Heinze, G., Wallisch, C., and Dunkler, D. Variable selection - A review and recommendations for the practicing statistician. *Biometrical Journal*. 2018;60(3):431-449. <https://doi.org/10.1002/bimj.201700067>
5. Kass, R. E., and Raftery, A. E. Bayes Factors. *Journal of the American Statistical Association*. 1995;90(430):773-795. <https://doi.org/10.1080/01621459.1995.10476572>
6. Hodson, T. O. Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geoscientific Model Development*. 2022;15:5481-5487. <https://doi.org/10.5194/gmd-15-5481-2022>
7. scikit-learn developers. `sklearn.metrics.r2_score` documentation. <https://sklearn.org/stable/modules/generated/sklearn>